

Telco Customer Churn Analysis

Customer Segmentation · PCA · Revenue at Risk

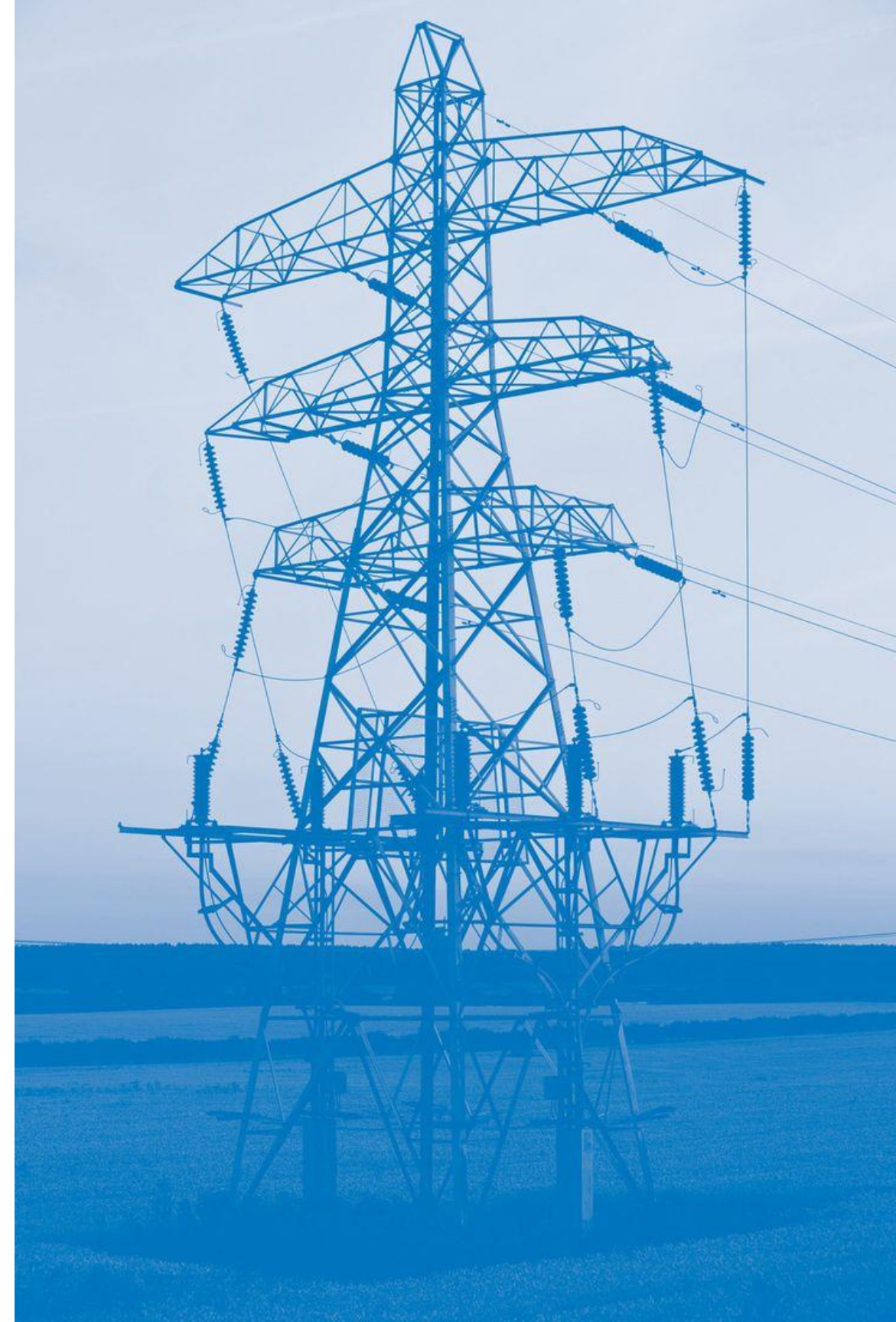
Business Value Through Integrative Analytics | B - MKT - 1

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Business Context & Data Overview

7,043	26.5%	54	4
Total Customers	Overall Churn Rate	Original Features	Clusters Identified
Comprehensive dataset spanning full customer base	Significant retention challenge requiring strategic intervention	Cleaned and engineered down to key analytical variables	Distinct customer segments via K-Means clustering

Key Churn Drivers from Exploratory Analysis

Month-to-Month Contract	Fiber Optic Internet	New Customers (<1 Year)
45.8% churn rate — lowest switching cost creates vulnerability	40.7% churn rate — high-value service with competitive alternatives	47.4% churn rate — fragile early-stage relationships

📌 Top churn reason: Competitor had better devices / made better offer (841 cases reported)

Feature Engineering

Three composite features reduce noise and add analytical value beyond raw variables:

Feature	Formula	Business Rationale
Service Bundle Count	Sum of 8 binary service flags (0–8)	Captures product depth and switching cost
Tenure Segment	Binned: New (0–12m), Developing (13–36m), Established (36m+)	Non-linear churn pattern — early stage is most fragile
Engagement Index	Sum of standardized referrals, bundle count, tenure	Composite loyalty signal across multiple dimensions

These engineered features enable more sophisticated segmentation and predictive modeling.

Cluster Analysis: Methodology

Four-Step Process

01

Feature Selection

7 clustering features: tenure, monthly charge, CLTV, satisfaction score, referrals, bundle count, engagement index

02

Standardization

StandardScaler applied to prevent high-magnitude variables (CLTV, charges) from dominating distance calculations

03

K Selection

Elbow method + Silhouette score evaluated K=3–8 ; K=4 selected for balance of fit and interpretability

04

K-Means Clustering

K=4, n_init=20, random_state=1 for stability and reproducibility

K Selection Results

Elbow Point	Silhouette Score
K=3	0.218 (Under-segments)
K=4 ✓ Selected	0.231
K=5	0.224 (Marginal gain)
K=6	0.219 (Over-segments)

Cluster Sizes

- Early High-Risk: 2,177 (30.9%)
- High-Value Bundled: 1,684 (23.9%)
- Loyal Advocate: 1,156 (16.4%)
- Stable Low-Usage: 2,026 (28.8%)

Cluster Profiles: Business Interpretation

Cluster 0: Early High-Risk Customers

Churn Rate: 67.3%	Newest customers, lowest engagement.
Avg CLTV: \$3,813	
Avg Tenure: 13.4 months	Highest churn.
Bundle Count: 3.2 services	

Cluster 1: High-Value Bundled Customers

Churn Rate: 15.9%	Longest tenure, highest CLTV.
Avg CLTV: \$5,025	Loss of even a small share
Avg Tenure: 54.2 months	creates large revenue impact.
Bundle Count: 5.7 services	

Cluster 2: Loyal Advocate Customers

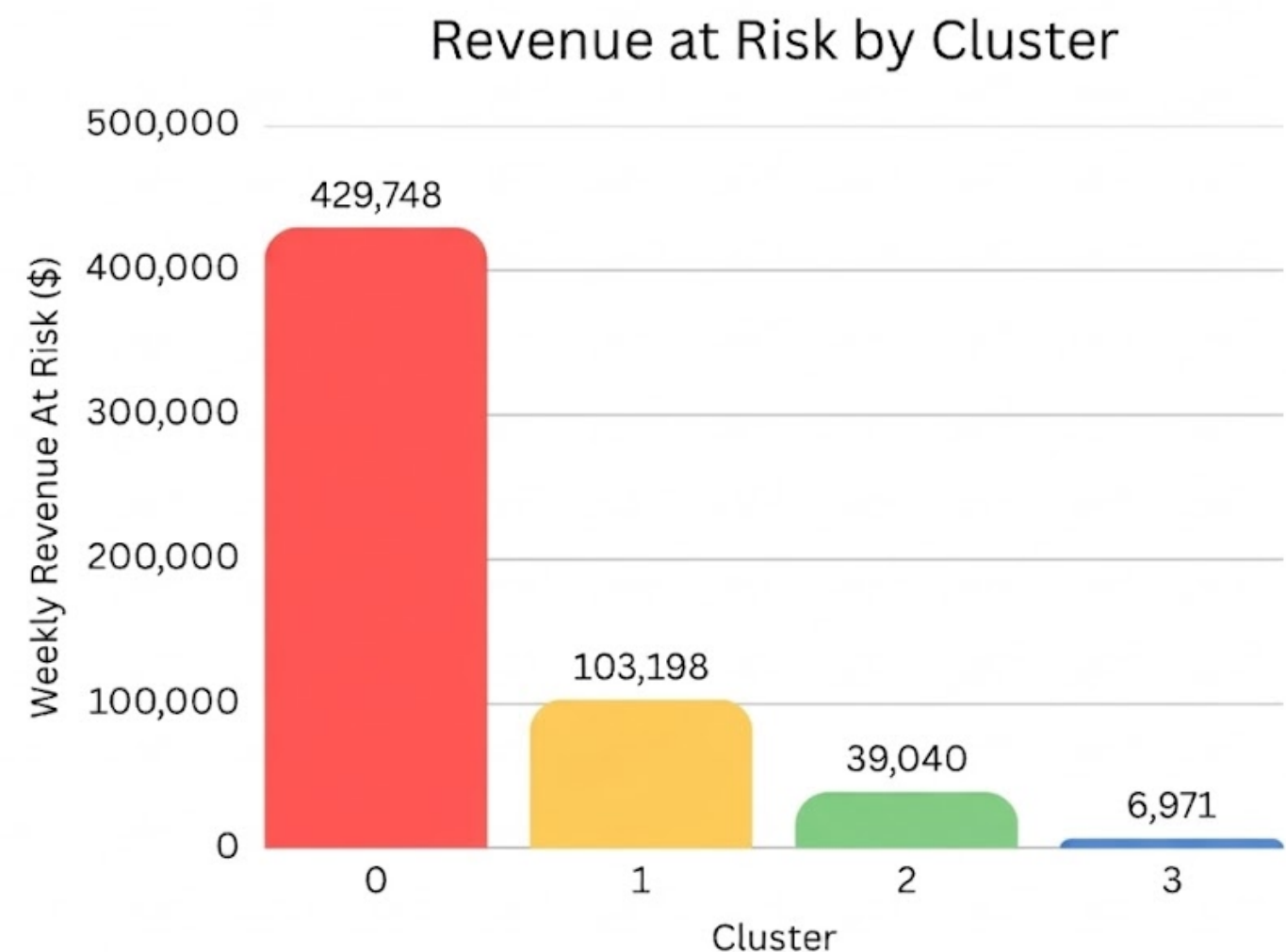
Churn Rate: 1.6%	Lowest churn, highest referral activity. Stable
Avg CLTV: \$4,770	base with indirect
Avg Tenure: 51.2 months	advocacy value.
Bundle Count: 4.0 services	

Cluster 3: Stable Low-Usage Customers

Churn Rate: 5.8%	Low spend, low engagement, but stable. High satisfaction
Avg CLTV: \$4,301	despite minimal service
Avg Tenure: 24.0 months	usage.
Bundle Count: 0.8 services	

Revenue at Risk Analysis

📄 **Weekly Revenue at Risk Formula:** $(\text{Segment Size} \div 13) \times \text{Churn Rate} \times \text{Average CLTV}$



Key Insight: Retention decisions must weigh **BOTH** churn probability **AND** customer value — not either alone. Cluster 0 represents highest priority despite moderate CLTV due to extreme churn rate.

PCA: Dimensionality Reduction & Validation

65.9%

PC1 + PC2 Variance

Explains over two-thirds of total variance in 2D space

97.1%

PC1–PC5 Variance

Five components capture nearly all information

Four clusters show visible structure in 2D PC space — not perfectly separated but distinct groupings exist.

PC1 — Customer Maturity

Primary Drivers:

- Engagement Index (+0.553)
- Service Bundle Count (+0.454)
- Tenure in Months (+0.452)

Represents customer lifecycle progression and product adoption

PC2 — Customer Experience

Primary Drivers:

- Satisfaction Score (+0.569)
- Monthly Charge (−0.525)
- Number of Referrals (+0.398)

Captures value perception and advocacy behavior

Conclusion: Clusters differ along interpretable business dimensions — not arbitrary groupings.

Key Takeaways & Managerial Implications

1 Early stage is the most fragile

New customers (<12 months) churn at 47.4%. Early relationship investment is **critical**.

2 Segmentation beats one-size-fits-all

Four distinct churn mechanisms exist. A homogeneous retention strategy leaves value on the table.

3 Risk = Probability × Value

Cluster 0 is top priority not just for high churn rate, but because it generates **\$429K in weekly revenue at risk**.

4 Protect the high-value base

Cluster 1 (High-Value Bundled) represents highest CLTV. Even moderate churn creates disproportionate loss.

Strategic Priority: Prioritize Early High-Risk customers first, then protect High-Value Bundled customers.

Next Steps: Predictive Modeling & Evaluation



Predictive Modeling

Use engineered features (service bundle count, tenure segment, engagement index, cluster assignment) as candidate predictors for churn classification.

⚠️ Exclude post-hoc variables (churn category, churn reason) to prevent data leakage.



Segment-Specific Strategy

Develop targeted retention interventions by cluster — early onboarding incentives for Cluster 0, loyalty programs for Cluster 1.

⚠️ Leverage cluster labels as actionable business segments.



Model Evaluation

Use precision-recall tradeoff appropriate for imbalanced churn data (26.5% positive rate). Evaluate AUROC and business-weighted metrics.

⚠️ Consider cost-sensitive learning to weight false negatives appropriately.

Thank You

Questions? Discussion welcome.

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